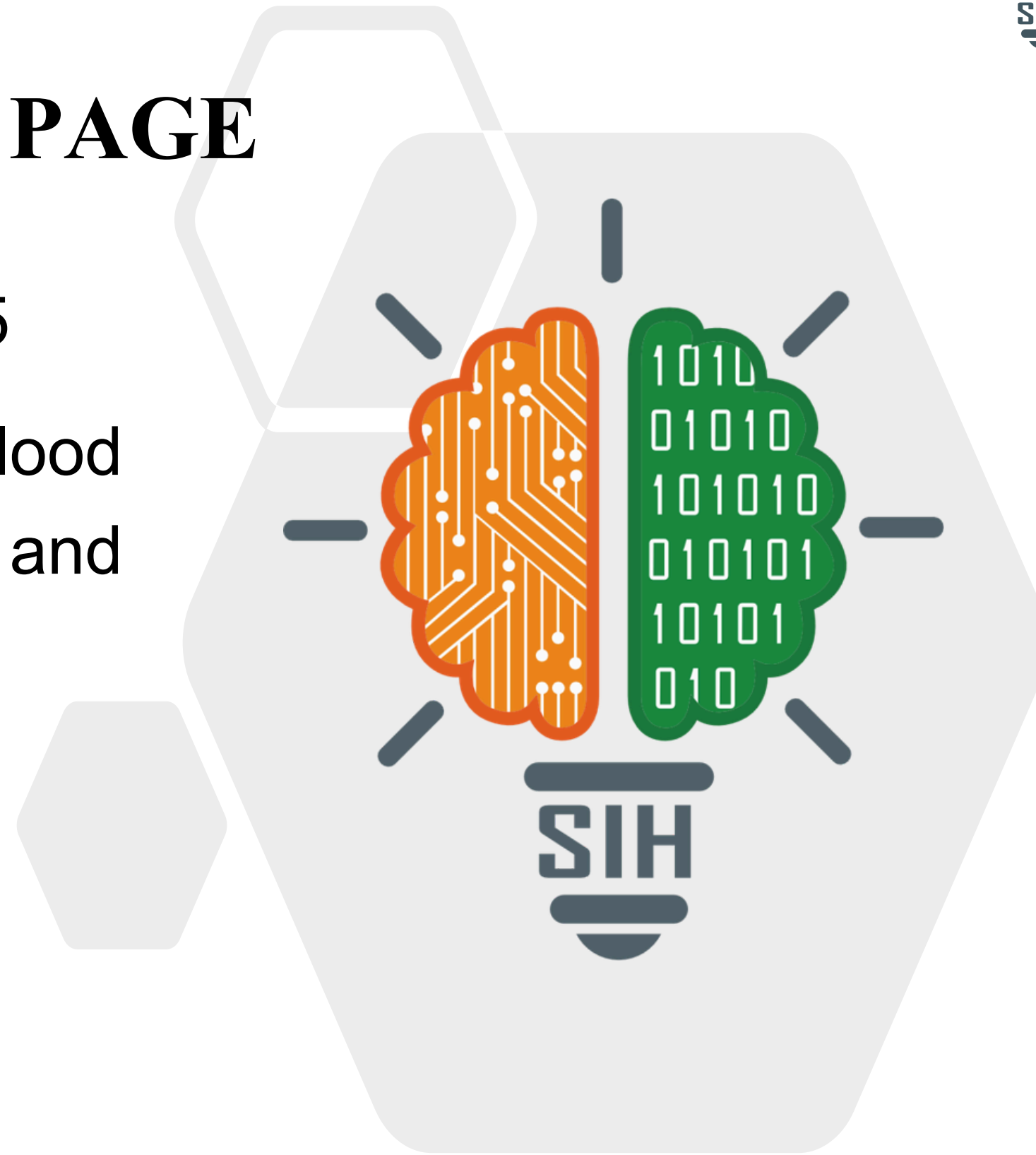


TITLE PAGE

- **Problem Statement ID** – SIH26085
- **Problem Statement Title**- Urban Flood Nowcasting System (Drainage and Rainfall Coupling)
- **Theme**- Disaster Management
- **PS Category**- Software
- **Team ID**-
- **Team Name** - 5D BEINGS

Demo Link : <https://5d-beings-beta.vercel.app/>



AI-Powered Urban Flood Decision Support System



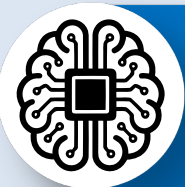
PROPOSED SOLUTION

-  **Predict**
Physics-Informed ML + Hydrodynamics
→ Water depth (cm), 0–3 hrs ahead
-  **Fuse**
IMD Radar + ISRO CartoDEM (10m) +
Storm-Drain Network
-  **Respond Faster**
<35 ms predictions → Real-time flood
assessment, replacing multi-hour
simulations

How It Addresses the Problem

- Eliminates Region-Wide Ambiguity:**
Traditional rain alerts ('65mm rain') fail to
specify flooded streets. Our system
outputs exact depth (68cm at Dwarka
Mor, 110cm at Kakrola Underpass).
- Flood-Safe Emergency Routing:**
Dynamically penalizes flooded roads
(W=999) to reroute ambulances & traffic
via dry detours (Pankha Road).

Innovation and Uniqueness

-  **AI-Based Flood Prediction**
-  **Interactive GIS Dashboard**
-  **Multi-Level Early Warning System**
-  **Drainage Network Analysis**
-  **Safe Route Recommendation**

TECHNICAL APPROACH

Data Ingestion

- Open-Elevation DEM (5x5 grid) - fetch_open_dem.py
- Open-Meteo radar & precipitation - fetch_live_radar.py
- OSM storm-drain vectors (Overpass) - fetch_osm_drainage.py
- Historical Excel rainfall & vashu ground-truth CSV

Feature Engineering

- 7 PS-PDF modules: radar reflectivity, infiltration, runoff, pipe fullness, depth, hazard index, navigation penalty - data_pipeline.py

Flood-Safe Routing

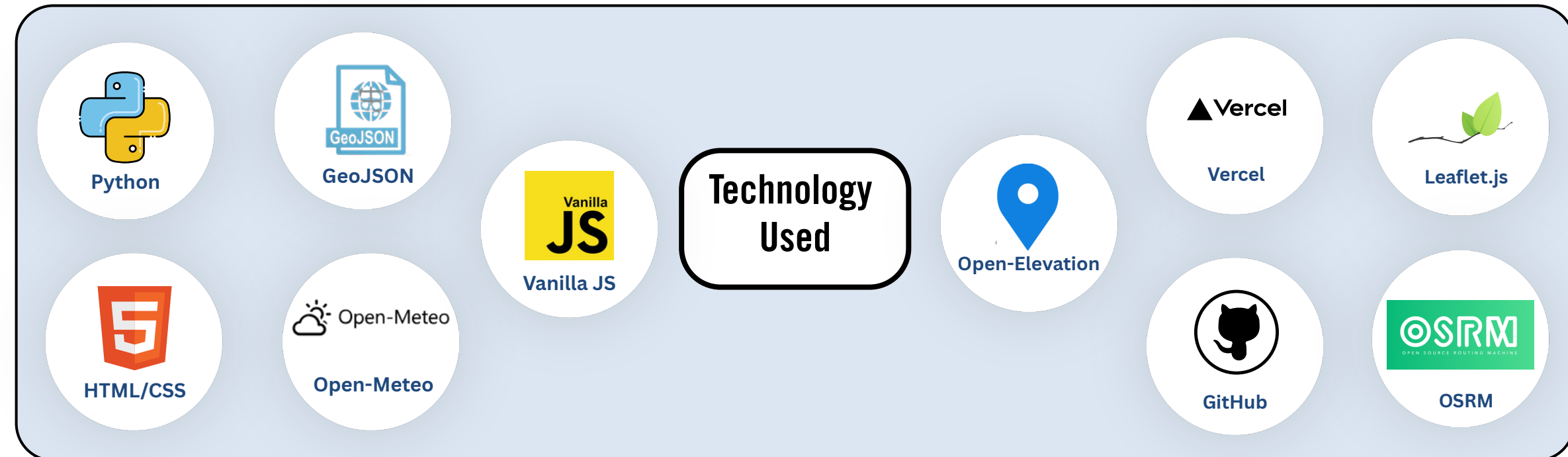
- OSRM routing + dynamic flood-depth risk check
- Vehicle-type clearance thresholds (ped 10 cm ... ambulance 45 cm) - /api/v1/route/flood-safe

API & Backend

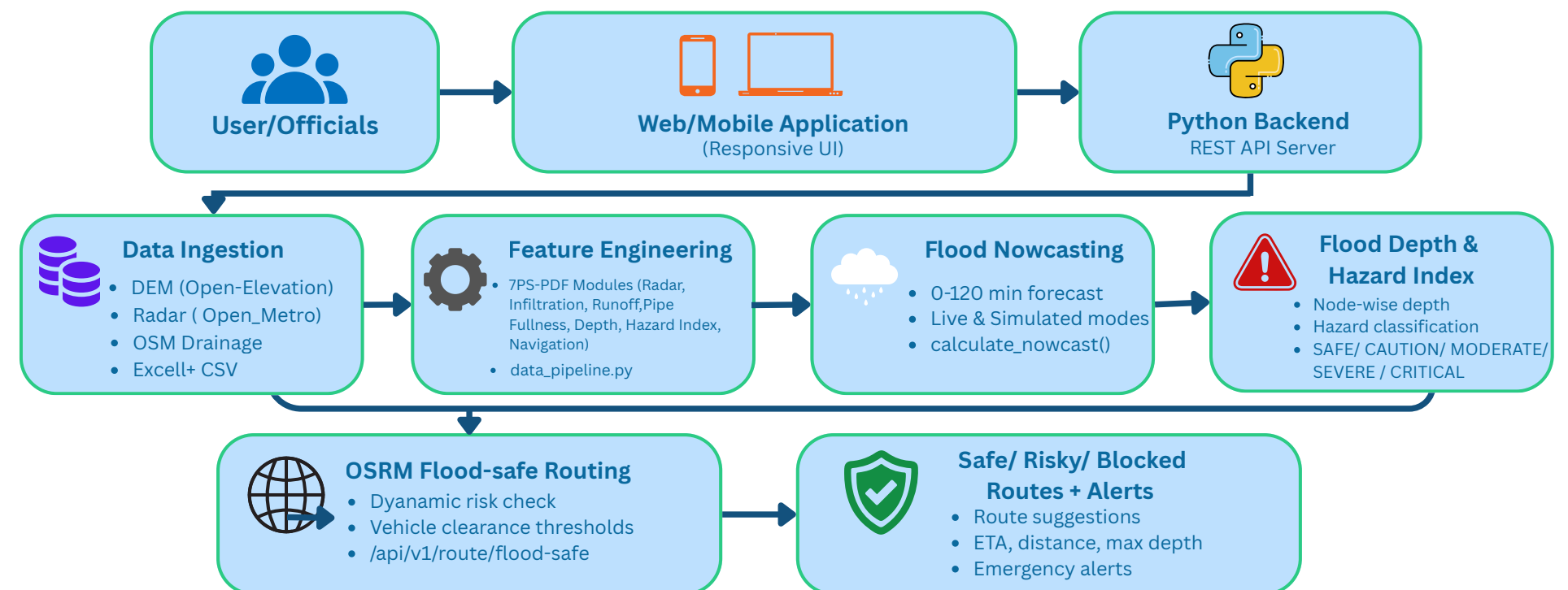
- Python HTTP server exposing /api/v1/* endpoints
- Swagger UI (/docs) for interactive API docs

Deployment & Reliability

- Local dev server (port 8081) + Vercel configuration (vercel.json)
- Offline fallback: synthetic DEM & simulated nowcast when external APIs fail



SYSTEM ARCHITECTURE



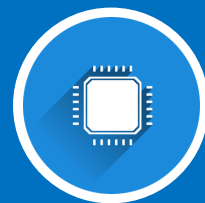
Technical Feasibility



- **Proven Data Grounding:** Trained on 15,264 hourly records & validated against 30 ground-truth flood points.



- **High Performance:** Achieved 0.055 cm MAE, 0.078 cm RMSE, 1.984 R² Score, & 98% Surcharge Accuracy.



- **Low Compute Cost:** Lightweight surrogate runs on standard CPUs without GPU requirements.



Potential Challenges And Risks

Risk 1 !
Radar Feed Latency

Risk 2 !
Incomplete Drain GIS Data

Risk 3 !
High Concurrency

Strategies

Fallback to Open-Meteo live rainfall API automatically.

Auto-inferred pipe slopes using ISRO Bhuvan 10m DTM.

Edge caching (1-year immutable rules) & script deferral.

Impacts

Reduces untreated sewage backflow



Environmental

Improves public safety by warning about water depths.



Social

Reduces losses caused by vehicle and pavement damages.



Economical



Benefits

Helps minimize water contamination



Environmental

Helps save lives and enables safer movements



Social

Enables faster emergency response and proactive deployment



Economical

1. India Meteorological Department (IMD) Data Portal: <https://dsp.imdpune.gov.in> (Palam Radar Station 28.5645° N, 77.1147° E)
2. ISRO Bhuvan Geo-Portal: <https://bhuvan.nrsc.gov.in> (CartoDEM 10m Urban Digital Terrain Model).
3. Irrigation & Flood Control Department Delhi (IFC): <https://ifc.delhi.gov.in> (Najafgarh Drain & Kakrola Regulator FSL Records).
4. US EPA SWMM 5.2 Model Specifications: Hydrodynamic directed graph friction equations (Manning's $n=0.013$).
5. Copernicus DEM Open Access: <https://dataspace.copernicus.eu> (Global 10m/30m Elevation Grid).